

Week 5 Homework 2 - RNA-seq analysis with EasyMultiProfiler-Web

Student: 郑烁瞳 (SUAT24000155) | Date: 2026-09-22 | App: EasyMultiProfiler-Web v9.0.4 Session: 1wtLICpulEfL2UusJulisZ9V, experiment rnaseq_week5, run bundle 20260922-185729. zip

1. Dataset identified in EasyMultiProfiler-Web/tests

File	Content
RNAseq_output.csv	raw integer count matrix, 24,394 genes (mouse symbols) x 24 libraries, first column feature
RNAseq_mapping.csv	sample metadata, 24 rows: SampleID (DHMR001-DHMR024) + Group

These are the only RNA-seq CSV files in the tests folder (the others are 16S and clinical). Design: 6 groups x 4 libraries - DMSO, DMSO+LIPUS, T4400, T4400+LIPUS, T3976, T3976+LIPUS (three treatments: vehicle, compound T4400, compound T3976; each +/- LIPUS ultrasound co-treatment).

2. How the analysis was run

The web UI's own HTTP API is the same code path the buttons call, so the analysis was driven through it (the UI was not opened because another local program already occupied port 8000):

1. POST /api/import/path with the two CSV files -> session 1wtLICpulEfL2UusJulisZ9V, experiment rnaseq_week5, 24 samples, 19,150 features retained after the app's import filter, sample overlap assay/metadata = 24/24, no warnings.
2. POST /api/workflows/rnaseq/run_all (one-click RNA-seq): PCA + scatter, DESeq2 differential (DMSO reference vs T4400), volcano, DEG/top-variance heatmaps, sample-correlation matrix, KEGG/GO enrichment. Results were packaged by the app into 20260922-185729. zip.
3. Five further DESeq2 contrasts through POST /api/analyze/differential (group_var = Group, subset_two_groups = true, DESeq2) with the full tables pulled from GET /api/analyze/diff_raw.
4. POST /api/github/sync to publish the results to the homework repository (see §7).

Reproducible scripts: run_contrasts.py (contrasts), verify_contrasts.py (independent check), hw2_stats.py (QC + panel statistics), hw2_figures.R (base-R figures).

3. Quality control (what the data actually looks like)

Observation	Number
Library size	20.5 - 40.4 million reads; 6 of 24 libraries at about half depth (DHMR001, 006, 016, 018, 019, 021)
Zero fraction per library	32.6 - 36.3 % of genes have zero counts (higher in the 6 shallow libraries)
Sample-sample Spearman correlation (log1p)	0.97 - 0.99 for all 552 pairs - no sample is globally broken
PCA (DESeq2 VST, 2,000 most variable genes)	PC1 = 71.7 %, PC2 = 9.4 % (figures/hw2_pca.png)

Observation	Number
PC1 by group (mean)	DMSO -9.7, DMSO+LIPUS -6.7, T4400 +14.4, T4400+LIPUS +16.1, T3976 -9.0, T3976+LIPUS -5.2
Single-sample immune responders (panel score > 1)	DHMR002 (DMSO), DHMR005 (DMSO+LIPUS), DHMR023 (T3976+LIPUS)

The dominant axis of variation (PC1) separates the T4400 arms from everything else, and the two strongest single-sample responders (DHMR005, DHMR023) also sit high on PC1. It is *not* a clean group separation: within each group one library - and in T4400 two libraries - carries a much stronger version of the same program (figure A of [figures/hw2_panel_heatmap.png](#)). This single-library structure is the most important QC finding, because it inflates the within-group variance and therefore decides which contrasts reach significance.

4. Differential expression results (DESeq2, padj < 0.05 and |log2FC| >= 1)

Contrast	Genes tested	Significant	Up	Down
DMSO vs T4400	16,757	238	172	66
DMSO vs T3976	16,606	1	0	1 (Krt19)
DMSO vs DMSO+LIPUS	16,630	0	0	0
T4400 vs T4400+LIPUS	16,730	0	0	0
T3976 vs T3976+LIPUS	16,584	4	4	0 (Kif26a, 3632451006Rik, Fxyd6, Tcap)
DMSO+LIPUS vs T4400+LIPUS	16,730	0	0	0

Only the T4400 arm (against vehicle) produces a substantial, threshold-passing response. Full tables: [tables/contrast_*.csv](#), [emp_run_bundle/tables/02_deseq2_raw.csv](#).

Top genes for DMSO vs T4400 (padj order): [Cep290](#) +1.28 (padj 8.9e-08), [Ptprv](#) -1.07 (4.2e-07), [Dcn](#) +1.45 (4.8e-06), [Il1a](#) +2.34 (6.2e-06), [Mest](#) -1.43, [Prdm1](#) +1.27, [Aldh1l2](#) +1.46, [Cc13](#) +5.12 (2.3e-05), [Erv3](#) +1.96, [Nrnl](#) +1.22, [Ero1l](#) +1.35, [Map3k8](#) +1.20, [P2rx3](#) +2.50, [Cxcl10](#) +1.33, [Cdkn1c](#) -1.25. Family content of the 238 genes: 7 [Serpina](#)*, 6 [Cc1](#)*, 5 [Mmp](#)*, 4 [Il](#)*, 3 [Cxcl](#)*, 2 [Saa](#)*, 2 [Kng](#)*, 2 [Mx](#)*, 2 [Ifit](#)*, 1 [Oas](#)*, 1 [Lcn](#)*, 1 [Try](#)*,

The induced program, and how consistent it really is

Mean counts per group for the program that dominates the result ([tables/panel_cpm.csv](#)):

Gene	DMSO	DMSO+LIPUS	T4400	T4400+LIPUS	T3976	T3976+LIPUS
Saa3	1,275	8,998	18,969	18,206	552	7,877
Lcn2	88	511	1,083	1,032	82	498
Mmp13	3.2	65.8	126.5	121.0	2.8	52.8

Gene	DMSO	DMSO+LIPUS	T4400	T4400+LIPUS	T3976	T3976+LIPUS
Ccl3	1.8	31.5	66.5	63.0	0.2	26.0
Kng1	21.5	229	472	454	10.5	219
Try5	9.0	164	304	348	3.8	148
Clec4e	1.0	18.5	34.5	36.8	1.0	16.0

Per-library counts show whether the shift is carried by all four replicates or by one library (examples, DMSO vs T4400): *Ccl3* 0/5/1/1 vs 12/23/134/97, *Mmp13* 0/11/2/0 vs 25/42/240/199, *Lcn2* 25/173/90/66 vs 178/316/2,044/1,792, *Saa3* 87/3,987/706/321 vs 3,134/7,097/35,870/29,773. For these genes every T4400 library is at or above the highest DMSO library, so the T4400 shift is a real group-level difference - but it is amplified two- to twenty-fold in one or two "responder" libraries, and the same panel flares up in single libraries of the other groups (DHMR005 in DMSO+LIPUS, DHMR023 in T3976+LIPUS), which is why the LIPUS-only and T3976 contrasts are empty.

4b. Figures

Figure 1 - PCA of the 24 libraries (DESeq2 VST, 2,000 most variable genes). PC1 (71.7 %) separates the T4400 arms from vehicle and T3976; the two single-sample responders (DHMR005, DHMR023) also sit high on PC1.

PCA of 24 RNA-seq libraries (DESeq2 VST, 2000 most variable

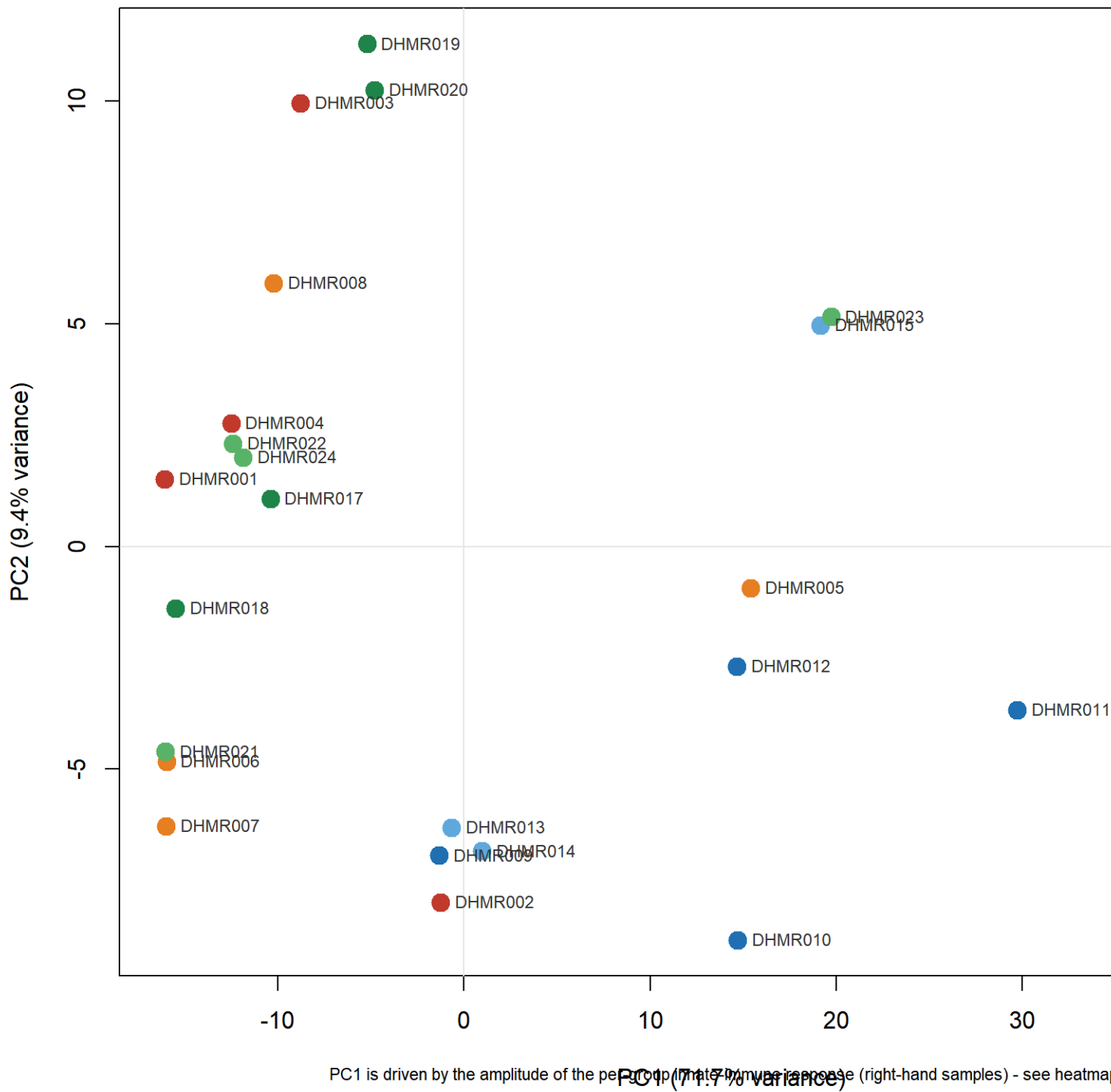


Figure 2 - A: the seven-gene induced panel (samples ordered by group, colour strip = group); one library per group - two in T4400 - carries an amplified version of the program. B: the 30 most variable genes with samples clustered by Spearman correlation; library size in brackets.

A. Induced innate-immune / acute-phase panel - samples ordered by group

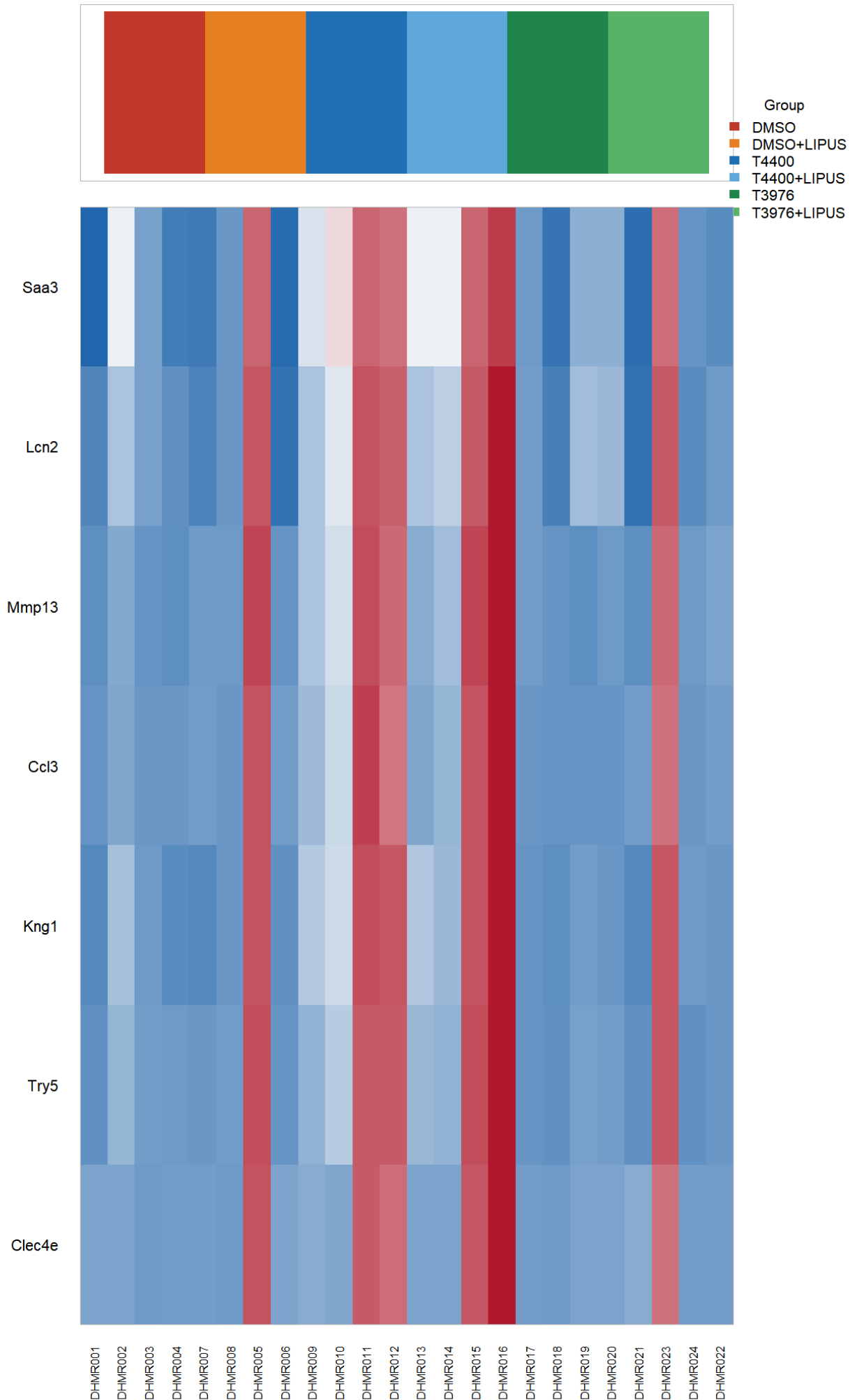


Figure 3 - Library size per sample; six libraries are at about half depth (dashed line 25 M).

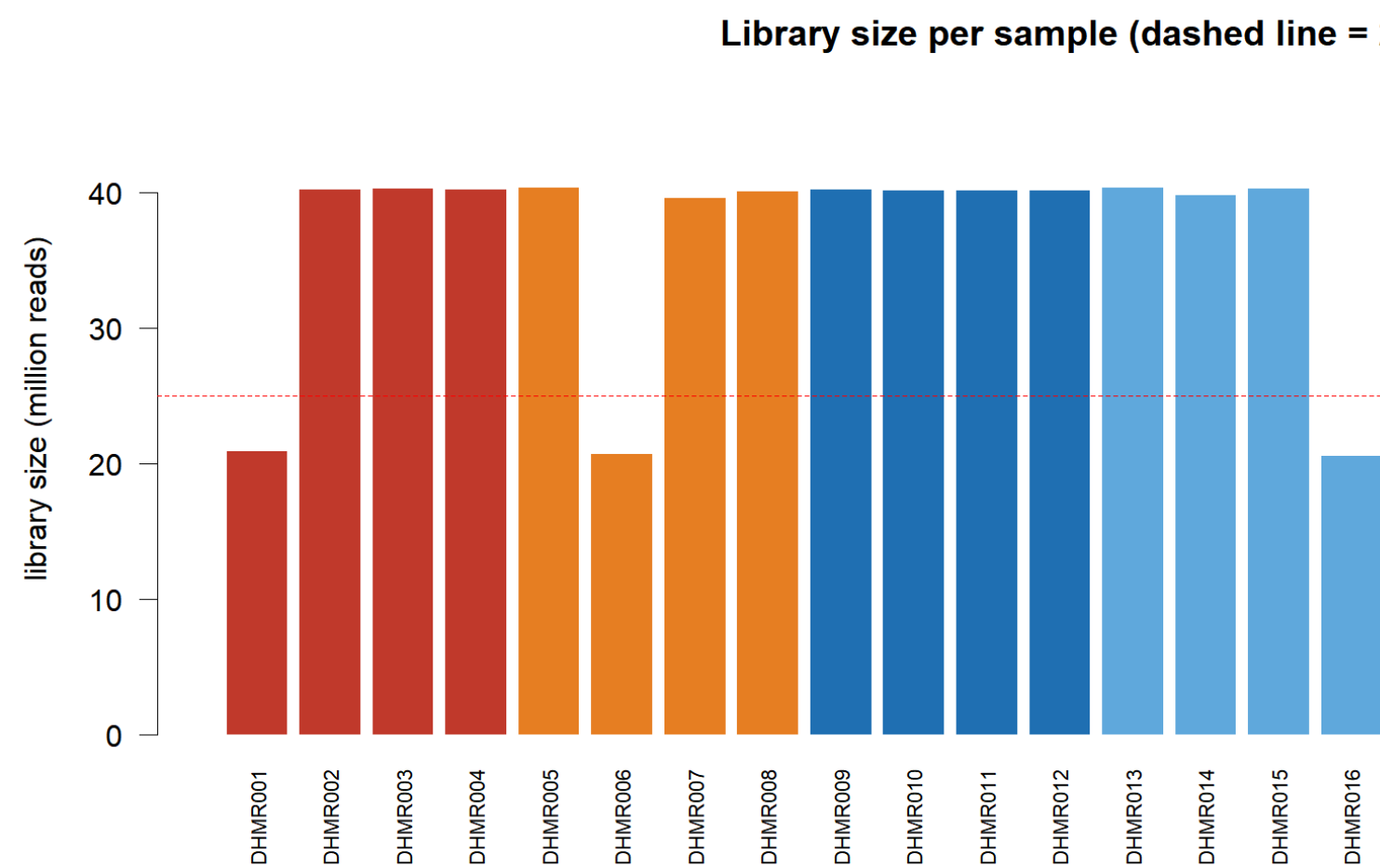
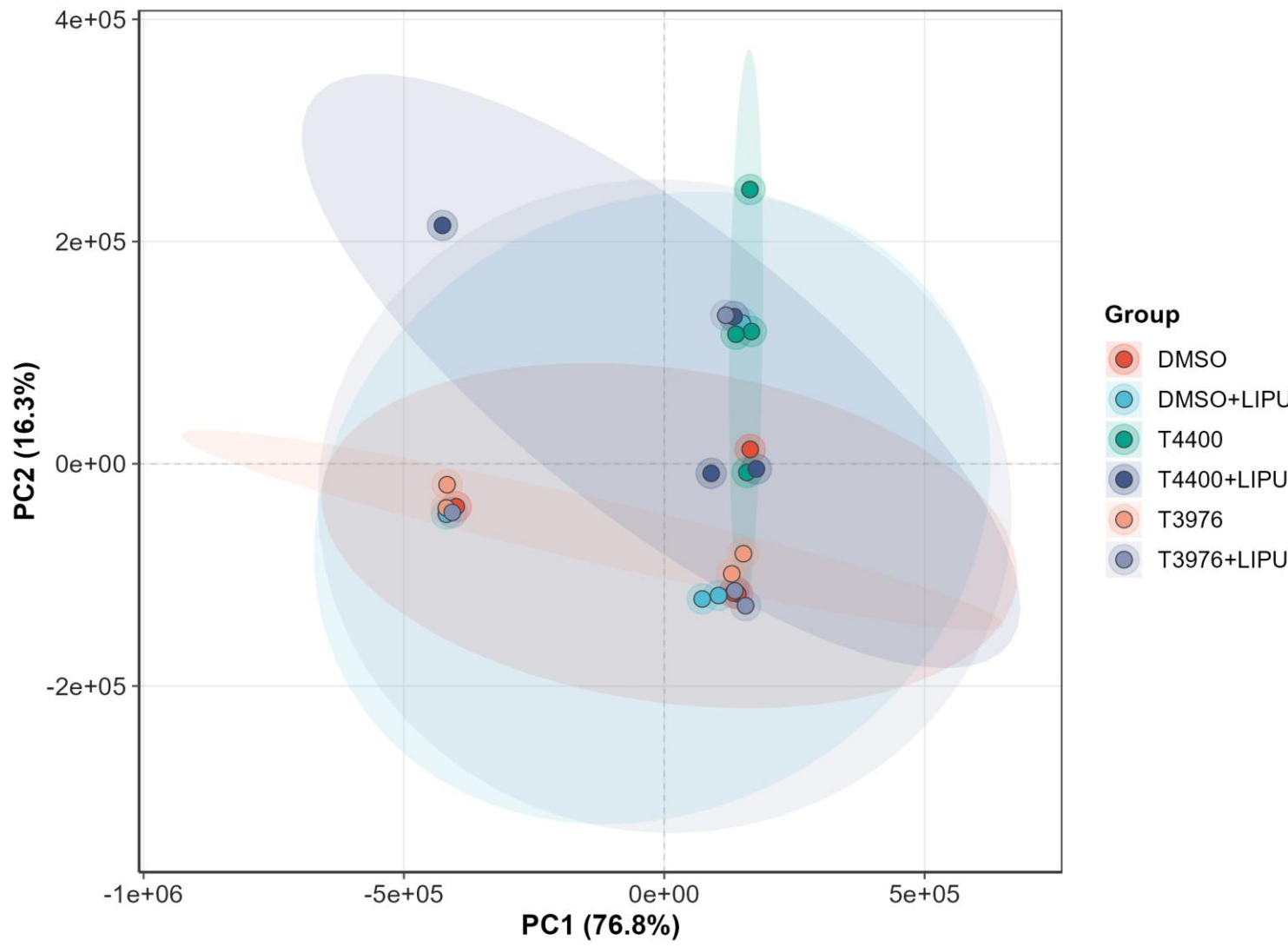


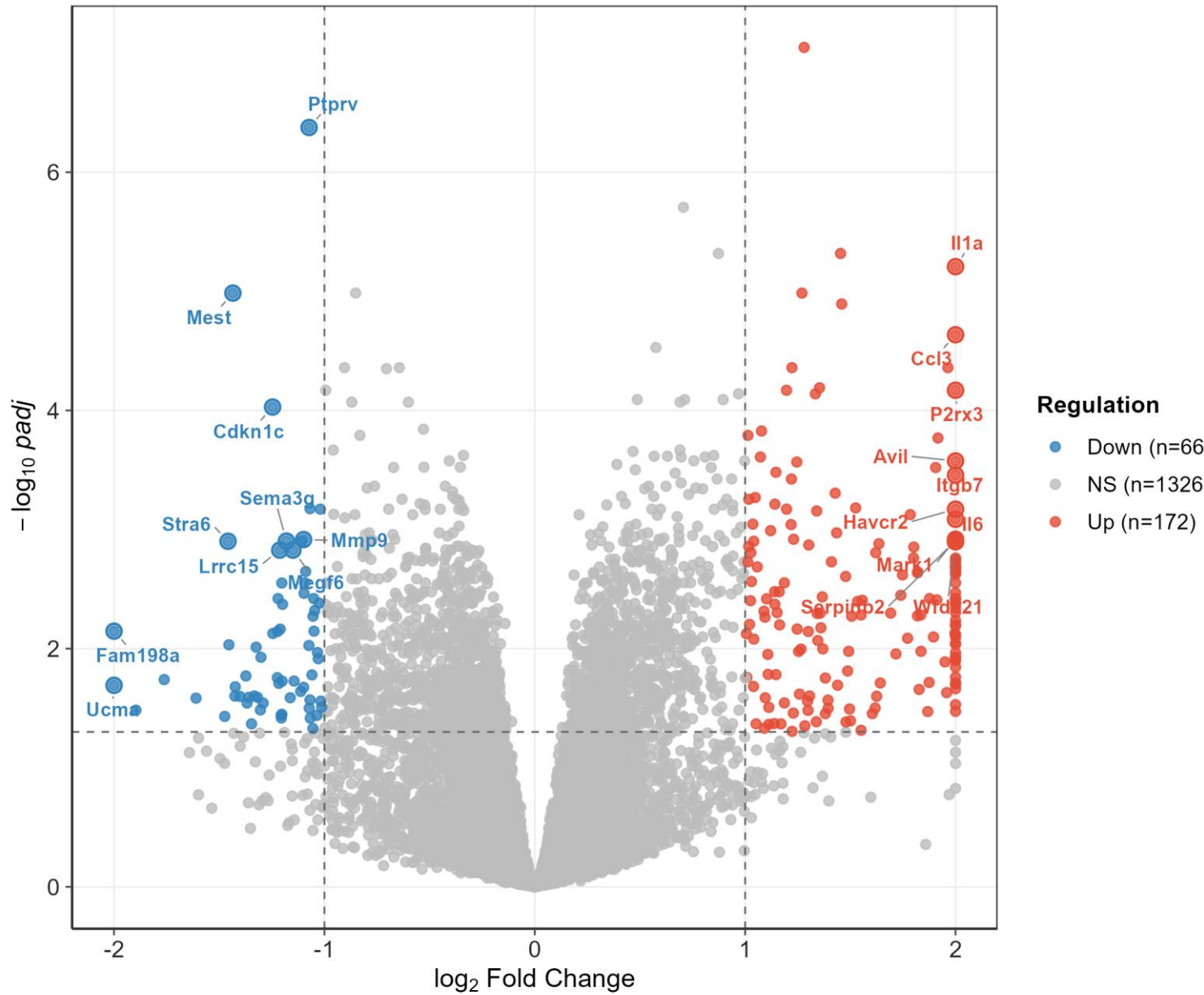
Figure 4 - The app's own figures from the one-click RNA-seq run (DMSO vs T4400): PCA scatter, volcano plot, top-variance heatmap and sample-correlation matrix.

PCA

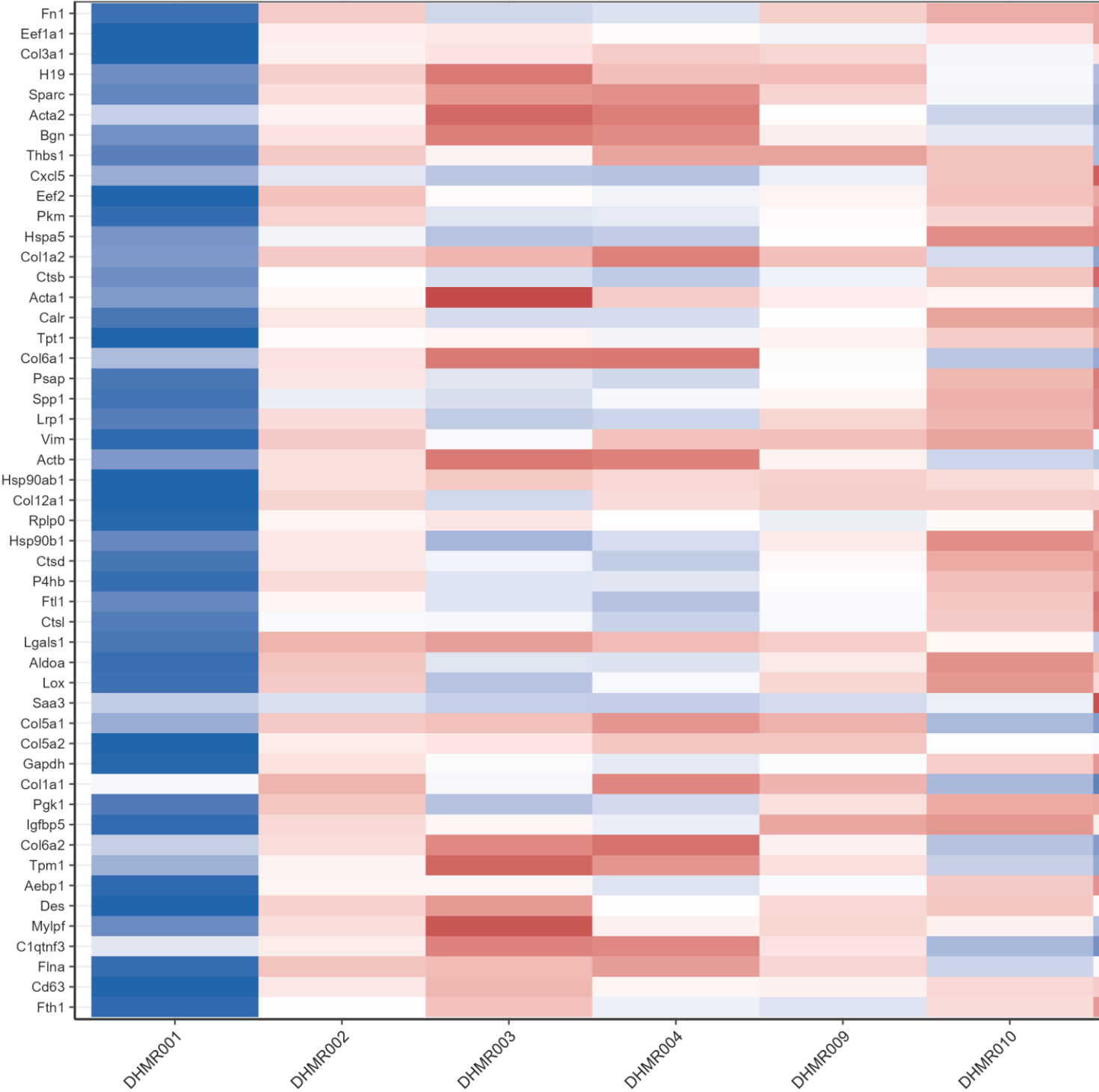


Volcano plot

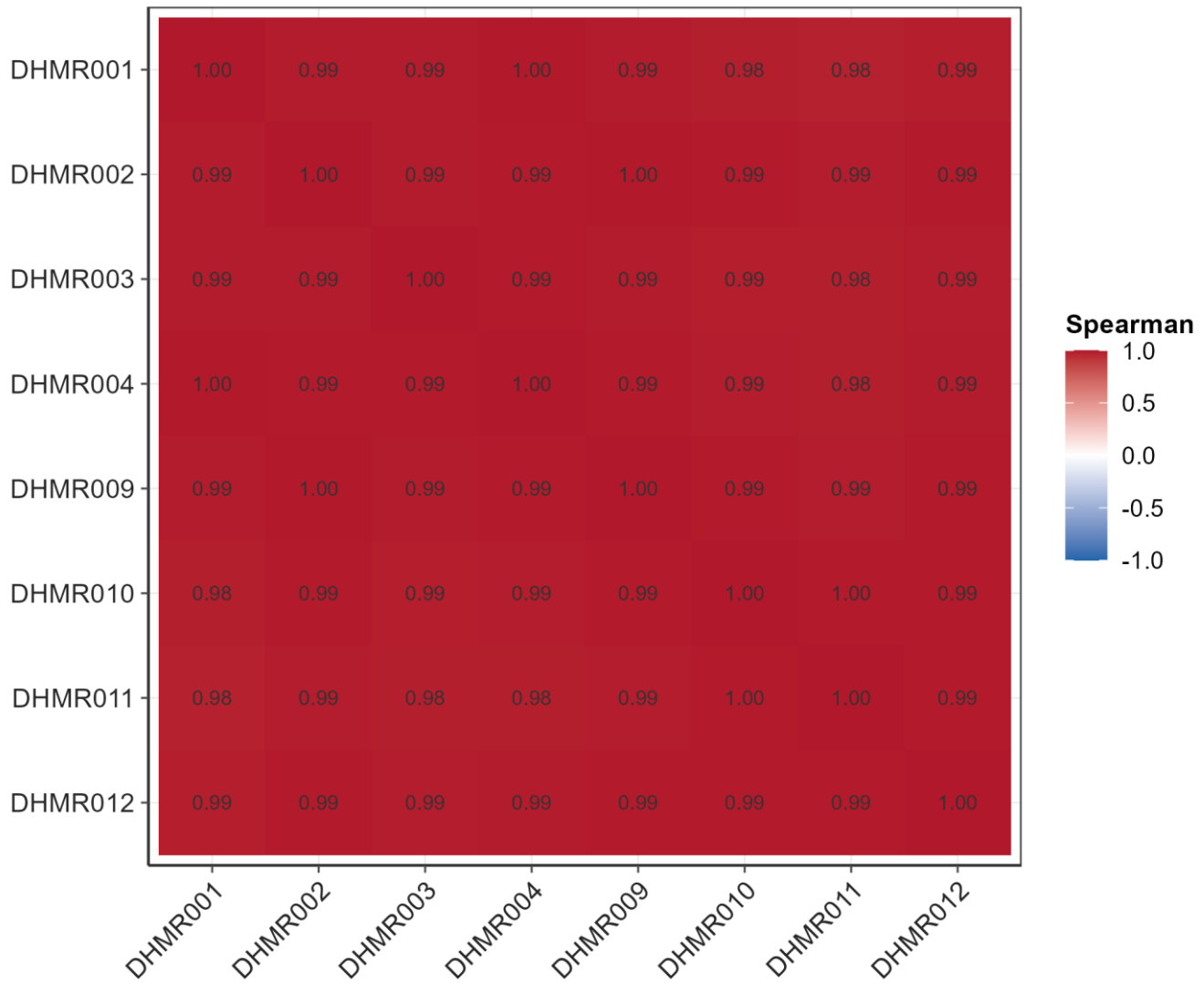
$|\log_2 \text{FC}| \geq 1, \text{padj} \leq 0.05$



Top 50 variable features (z-scored)



Sample-level correlation



5. Verification of the app's output (independent checks)

Check	Result
Contrast assignment (app log2FC vs my pseudo-bulk log2FC computed straight from the CSV)	Pearson $r = 0.944$ (DMSO vs T4400) and 0.951 (DMSO+LIPUS vs T4400+LIPUS) - the right samples were compared
Stored DESeq2 table internally consistent ($pvalue == 2 * pnorm(-abs(stat))$, BH monotone)	$r = 1.000$, zero BH violations
App DEG list <code>tables/04_deg_list.csv</code>	1,395 rows, but 1,157 of them are all-NA placeholder rows; only 238 pass $padj < 0.05$ and $ log2FC \geq 1$. The headline "1,395 DEGs" from the bundle is an over-count; 238 is correct
p-values in the JSON export	rounded to 4 decimals, so $p < 1e-4$ is exported as 0; $padj$ (used for every threshold here) is unaffected at $padj < 0.05$
Enrichment	KEGG and GO both failed with "OrgDb package <code>org.Mm.eg.db</code> for organism <code>mmu</code> is not installed", so pathway enrichment is not available in this run (logged in <code>emp_run_bundle/summary.txt</code>)

Check	Result
Library depth vs result	the two most extreme PC1 libraries include both a half-depth library (DHMR016) and a full-depth one (DHMR011), so depth alone does not explain the pattern

6. Biological interpretation

The T4400 arm shows a coordinated **innate-immune / acute-phase plus extracellular-matrix remodelling** response. Acute-phase and iron-sequestration genes (*Saa3* ~15x, *Lcn2* ~12x), chemokines and their receptors (*Ccl3* ~37x, *Ccl8*, *Cxcl1*, *Cxcl10*), interferon-stimulated genes (*Mx2*, *Oas**, *Ifit2*), matrix metalloproteinases and proteases with their inhibitors (*Mmp13* ~40x, *Try5* ~34x, *Serpinb2*, *Serpina3n*), the innate sensing lectin *Clec4e* (*Mincle*), *Il1a*/*Il6*, and the ECM protein *Dcn* all move in the same direction, together with cell-cycle arrest markers (*Cdkn1c*) and *Prdm1*. In functional terms this is a wound/inflammation-like transcriptional state: leukocyte chemoattraction, acute-phase protein secretion and matrix degradation, which is the response expected when cells are exposed to a pro-inflammatory or stress-inducing insult rather than a clean differentiation stimulus.

Whether that state is caused by the compound itself cannot be settled here. The same program appears, at lower amplitude, in one library of the vehicle + LIPUS group and one of the T3976 + LIPUS group, so the response looks like a cell-state switch that occurs in a subset of libraries in every arm, with T4400 pushing more libraries across the threshold. LIPUS alone (0 genes) and T3976 alone (1 gene) do not reproduce it, and adding LIPUS to either compound adds nothing detectable (4 genes for T3976+LIPUS vs T3976, 0 for T4400+LIPUS vs T4400), i.e. there is no add-on effect of ultrasound at this sample size.

7. Submission

- Results, figures, tables and scripts of this week are in this folder and were pushed to `zhashutiaoye-sketch/Bioinformatics_homework_Zheng-Shuotong` under week5/.
- `POST /api/github/sync` was used to publish the app's own run bundle for this session (the student account is bound to the repository above); the local sync history is returned by `GET /api/github/syncs`.

8. Limitations

- $n = 4$ per group, and one library per group carries an amplified version of the induced program, so within-group variance is dominated by single libraries; the T4400 result is therefore hypothesis-generating, not confirmed.
- The identities of T4400 / T3976 and the origin of the DHMR dataset are not documented in the `tests` folder, so the interpretation stays descriptive (no named pathway or target).
- No pathway enrichment was possible (`org. Mm. eg. db` missing), so the functional reading relies on gene-symbol families, not on a statistical over-representation test.
- 6 of 24 libraries are at half sequencing depth; DESeq2 size factors correct the overall level but power in those groups is lower.
- The DE call uses the app's raw `pvalue/padj` from the stored DESeq2 table; the app's exported CSV rounds p-values, which is why the counts in §4 were recomputed from the full-precision `padj` column.

9. AI use and verification (documented)

The analysis was executed by an AI assistant working on this machine; every number above comes from the app's own run or from the verification scripts in this folder, and both were re-run and inspected. Two AI-relevant corrections are recorded in §5: the app's DEG list over-count (1,395 -> 238) and the exported p-value rounding. The interpretation was written by the assistant and checked against the per-library counts reproduced in §4.